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BST vs. Hash Table.

In this work I made 2 structures in java: first is BST, second is Hash Table. BST is a tree while Hash table is a list of buckets.

I compared 3 actions for both of them: add, find and delete. For checking I used Benchmark with several tests. Tests checked every work by the correctness of termination. Also I checked searching for numbers that don't exist and all tests passed.

After that I compared all actions by the time. So I made a speed test. I used 100 numbers, then 1000 numbers and so on. The numbers were all different and in random order. Both BST and HT had a same numbers. I ran the test 5 times to warm up. Then I ran it 20 times and took the average. Time is in milliseconds.

About the results: when we have 10,000 numbers HT is much faster, for find and delete faster about 12 times and for insert 2 times. But with small numbers insert in BST is faster, however HT is faster for searching and deleting.

In conclusion, if you need to add and find and delete numbers very fast, use HT. HT is better for big data. However, BST keeps numbers in order, that HT cannot to do this, so if the order is important use BST.